Sinusoidal Model

From spectral peaks to time-varying sinusoidal tracks and additive resynthesis

Digital Signal Processing for Sound and Music · Topic 5

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Lecture roadmap

How the topic moves from spectral peaks to a working analysis/synthesis system

1 Model idea
time-varying sinusoids

2 Spectral peaks
windows, lobes, resolution

3 Peak detection
zero-padding + interpolation

4 Tracks
connect peaks across frames

5 Synthesis
main-lobe generation + OLA

6 Examples + Lab 5
inspect, listen, tune parameters

From STFT to a higher-level representation

The sinusoidal model keeps the prominent, interpretable spectral structure

- STFT keeps a full complex spectrum for every frame.
- Sinusoidal modeling keeps selected peaks and tracks.
- We lose some identity information, but gain abstraction and flexibility.

time-domain samples → STFT frames → spectral peaks → sinusoidal tracks

What changes in the representation?

- from one spectrum column to a list of peaks
- from frame-level values to continuous tracks
- from analysis plots to synthesis parameters

Core idea: find peaks, link peaks, synthesize tracks.

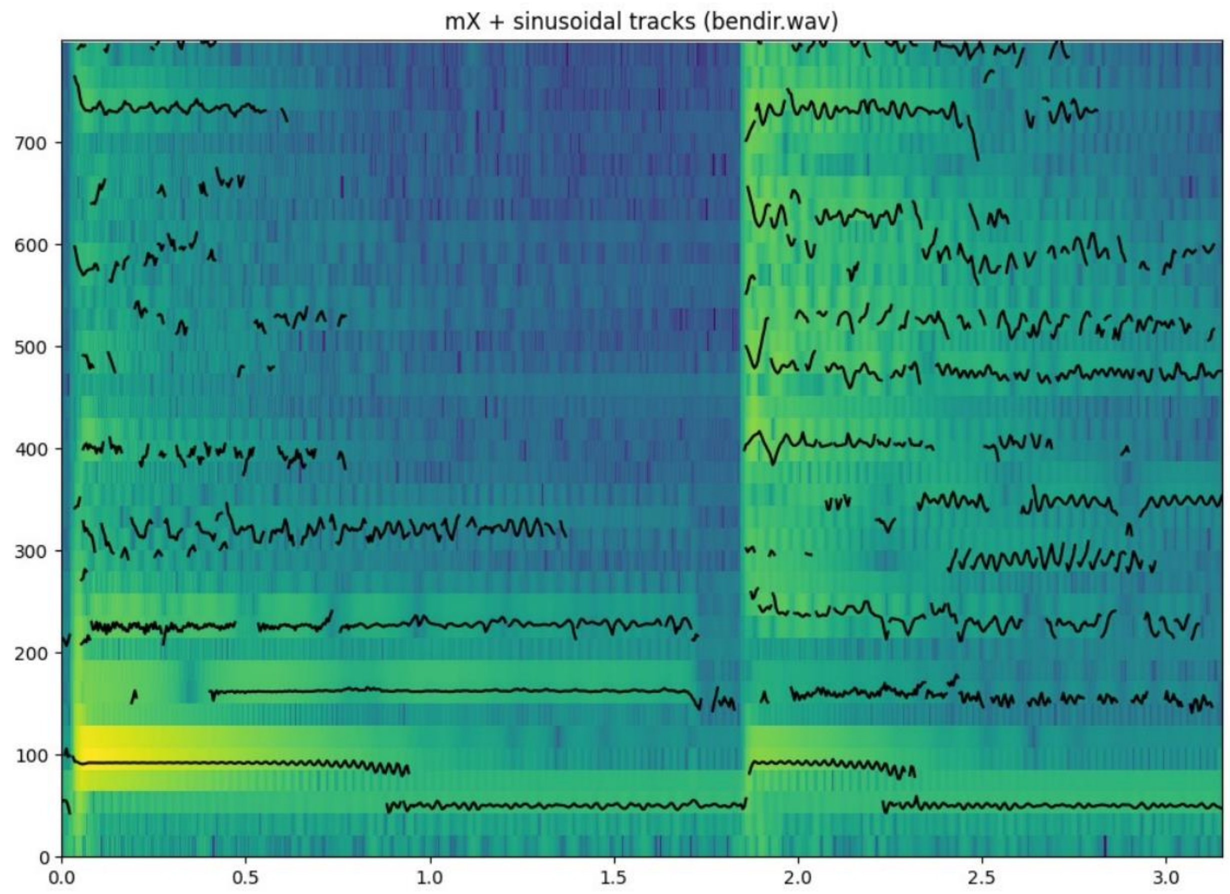
MODEL

Sinusoidal model

A sound is represented as a sum of time-varying sinusoids

$$y[n] = \Sigma_r A_r[n] \cos(2\pi f_r[n] n + \varphi_r[n])$$

- r indexes the sinusoidal component
- $A_r[n]$ is instantaneous amplitude
- $f_r[n]$ is instantaneous frequency
- $\phi_r[n]$ is phase



Tracks show frequency over time; magnitude and phase are also stored.

Computing the spectrum of a sinewave

Use the complete derivation to connect the time-domain cosine with its two shifted window spectra

$$x[n] = A\cos(2\pi k_0 n/N + \phi)$$

$$X[k] = A\sum_{n=0}^{N-1} w[n]\frac{1}{2}\left(e^{j2\pi k_0 n/N} + e^{-j2\pi k_0 n/N}\right)e^{-j2\pi kn/N}$$

$$= \frac{A}{2}\sum_{n=0}^{N-1} w[n]e^{j2\pi k_0 n/N}\, e^{-j2\pi kn/N} + \frac{A}{2}\sum_{n=0}^{N-1} w[n]e^{-j2\pi k_0 n/N}\, e^{-j2\pi kn/N}$$

$$= \frac{A}{2}\sum_{n=0}^{N-1} w[n]e^{-j2\pi(-k_0+k)n/N} + \frac{A}{2}\sum_{n=0}^{N-1} w[n]e^{-j2\pi(k_0+k)n/N}$$

$$= \frac{A}{2}W[k-k_0] + \frac{A}{2}W[k+k_0]$$

Key idea

- start from a finite cosine
- split it with Euler's formula
- collect the two complex exponentials
- each term becomes a shifted copy of W[k]
- result: two symmetric lobes at $\pm k_0$

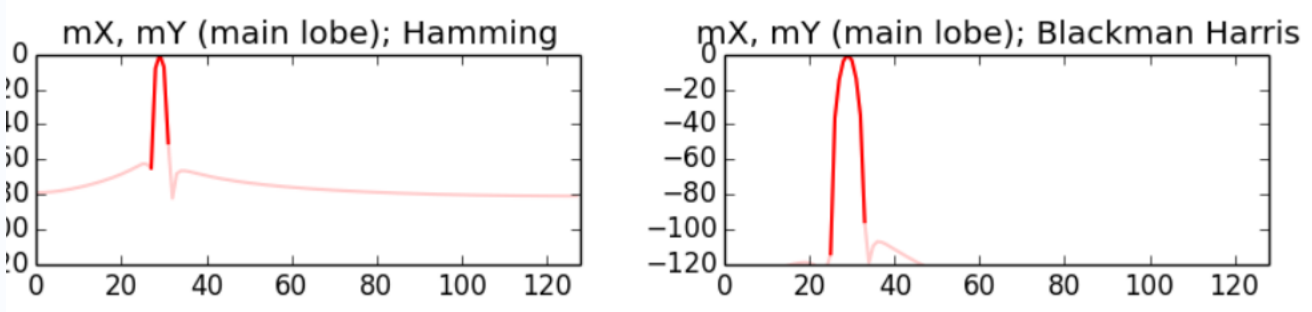
Windowing shapes every sinusoidal peak

The spectrum belongs to the windowed signal, not to an ideal infinite sinewave

Window main lobe and bin spacing

$$B\_f = B_s \cdot f_s \,/\, M$$
$$\Delta = |\, f_{k+1} - f_k \,|$$
$$M \geq B_s \cdot f_s \,/\, \Delta$$

B_s: main-lobe width in bins
f_s: sampling rate
M: window size
Δ: spacing between sinusoids



Different windows create different main-lobe widths and side-lobe levels.

The window spectrum sets the shape of every sinusoidal peak.

Window-size consequences

- larger M separates closer frequencies
- smaller M follows faster time changes
- mismatched lengths create leakage
- zero-padding samples the same spectral shape more densely

Choosing M is a modeling decision

A wrong window size can merge, smear, or distort sinusoidal components

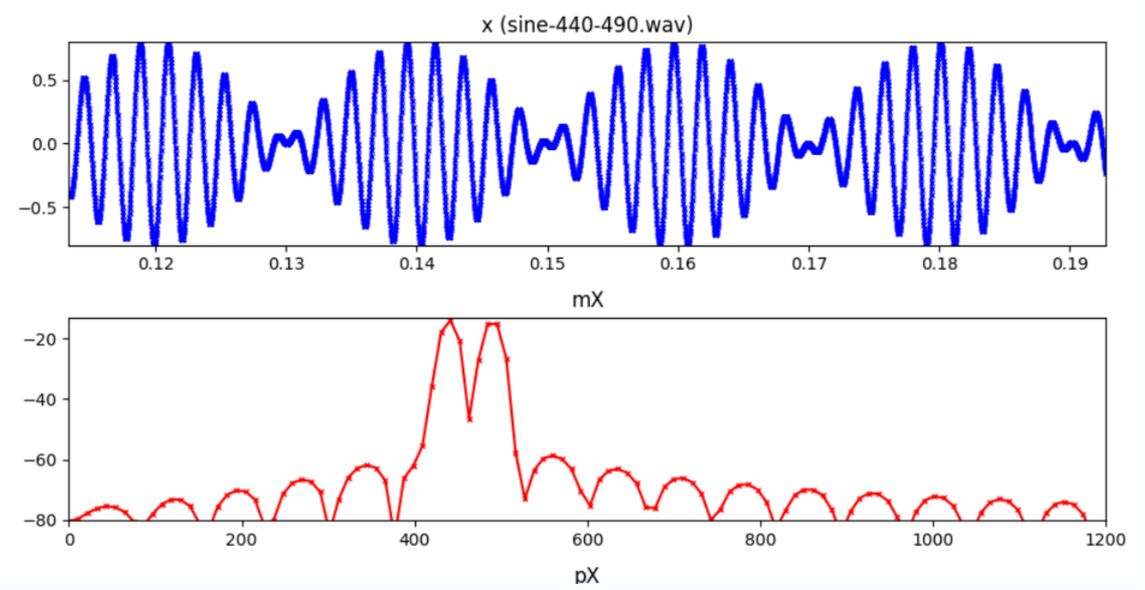
Example: 440 Hz and 490 Hz

Hamming window: $B_s = 4$
$f_s = 44100$ Hz
$\Delta = |490 - 440| = 50$ Hz

$$M \geq B_s \cdot f_s / \Delta$$
$$M \geq 4 \cdot 44100 / 50$$
$$M \geq 3528 \text{ samples}$$

M is chosen from the frequencies we want to separate.

Plot check: two resolvable peaks

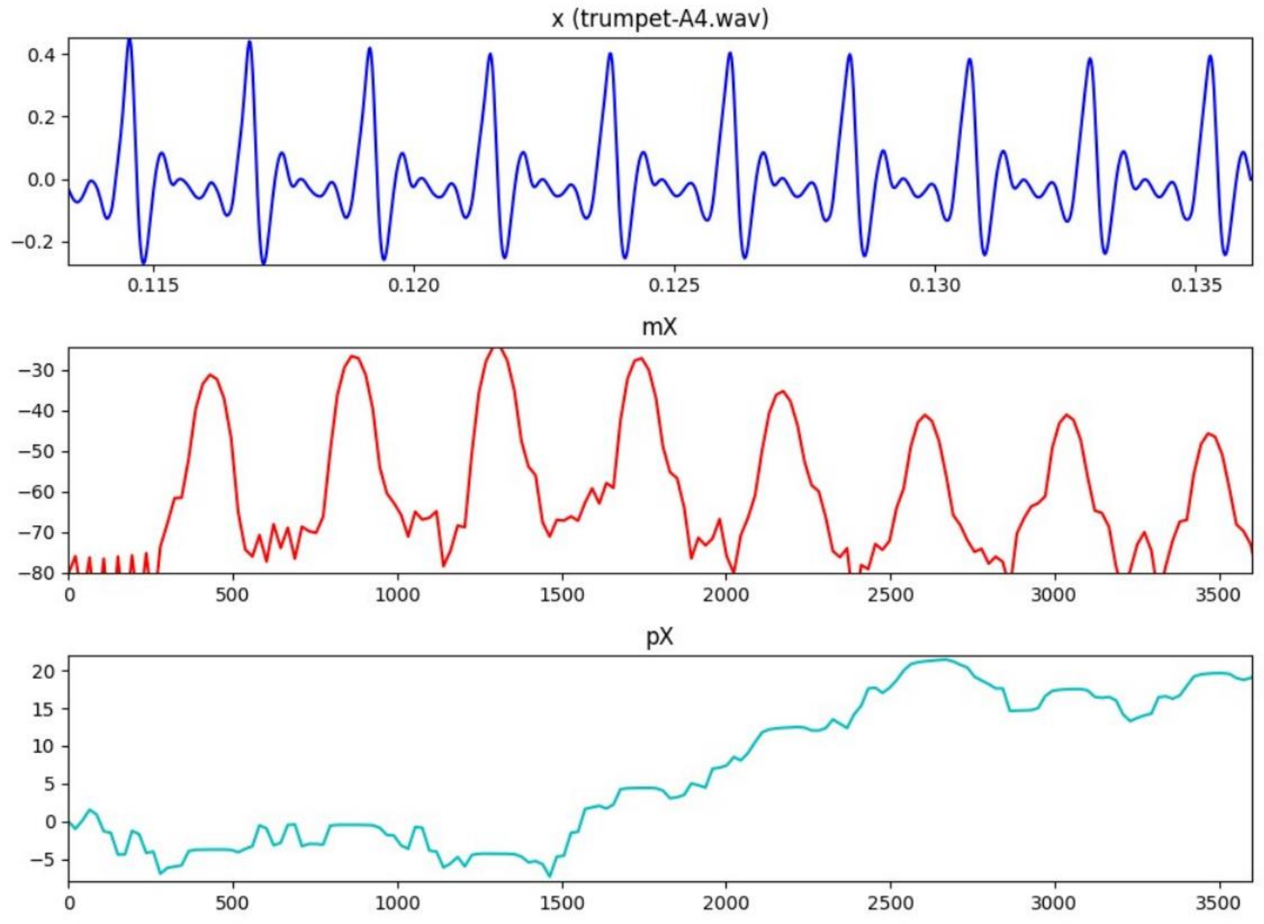


The waveform shows beating; the magnitude spectrum should show two separate peaks.

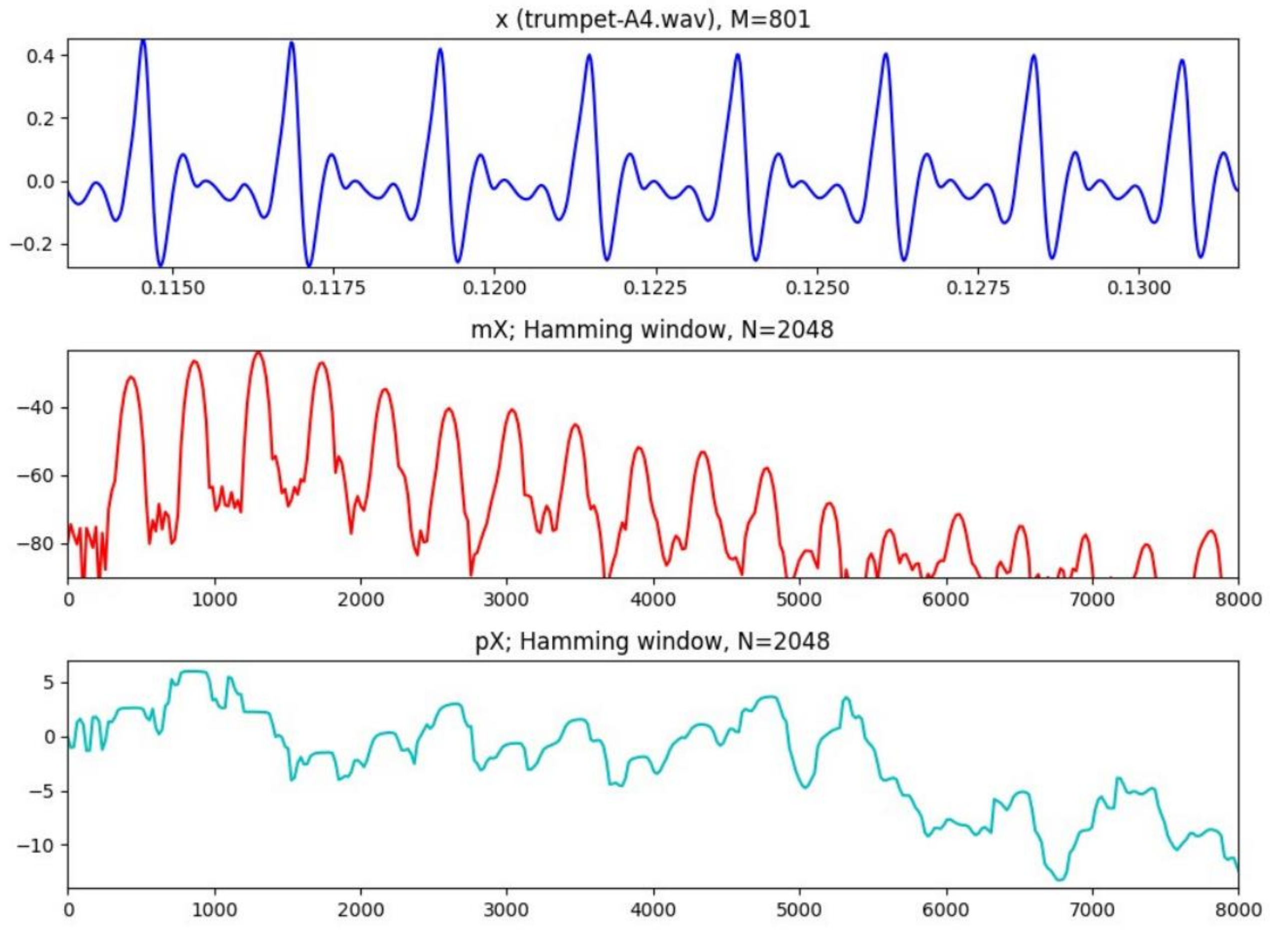
EXAMPLE

Real musical sounds contain many overlapping peaks

The sinusoidal model is useful when the important structure can be tracked as peaks



oboe: many peaks in a real sound



trumpet: spectral peaks and phase

Peak detection

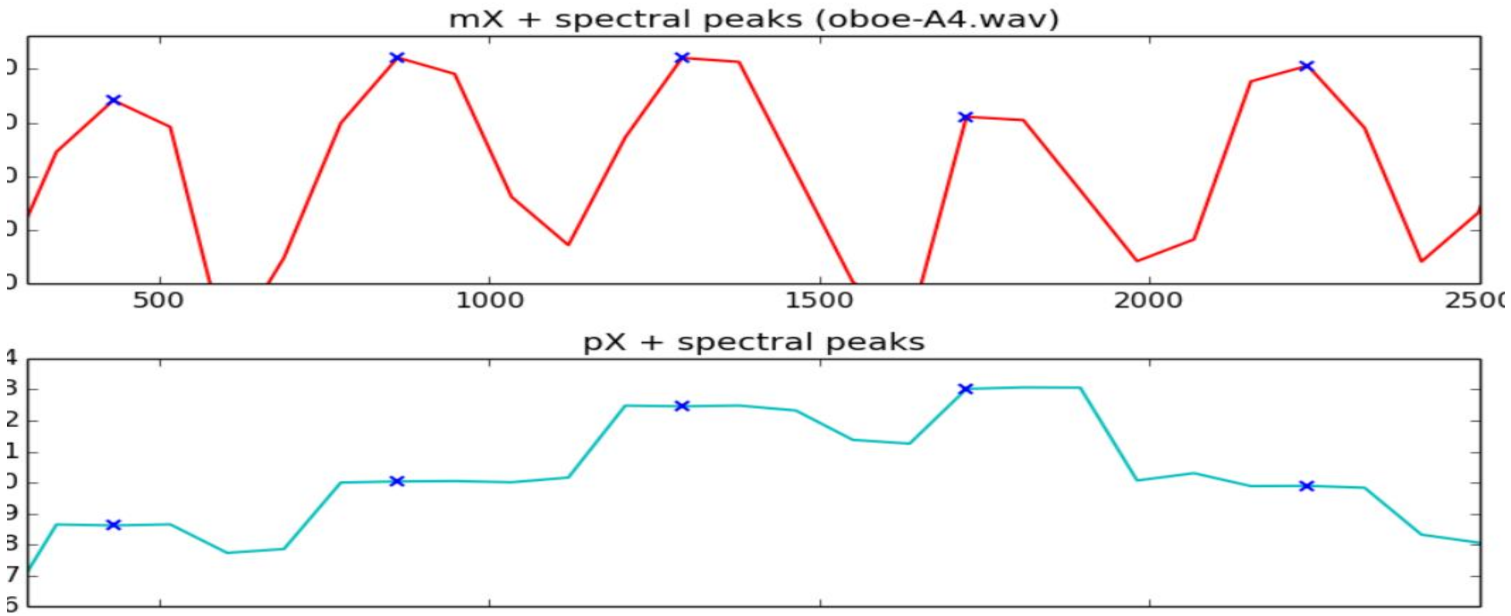
First identify local maxima above a threshold in the magnitude spectrum

Local-maximum test

peak at bin k if

$mX[k] > t$

$mX[k] > mX[k-1]$ and $mX[k] > mX[k+1]$



The plot now shows the full magnitude example without the previous crop.

Typical interpretation

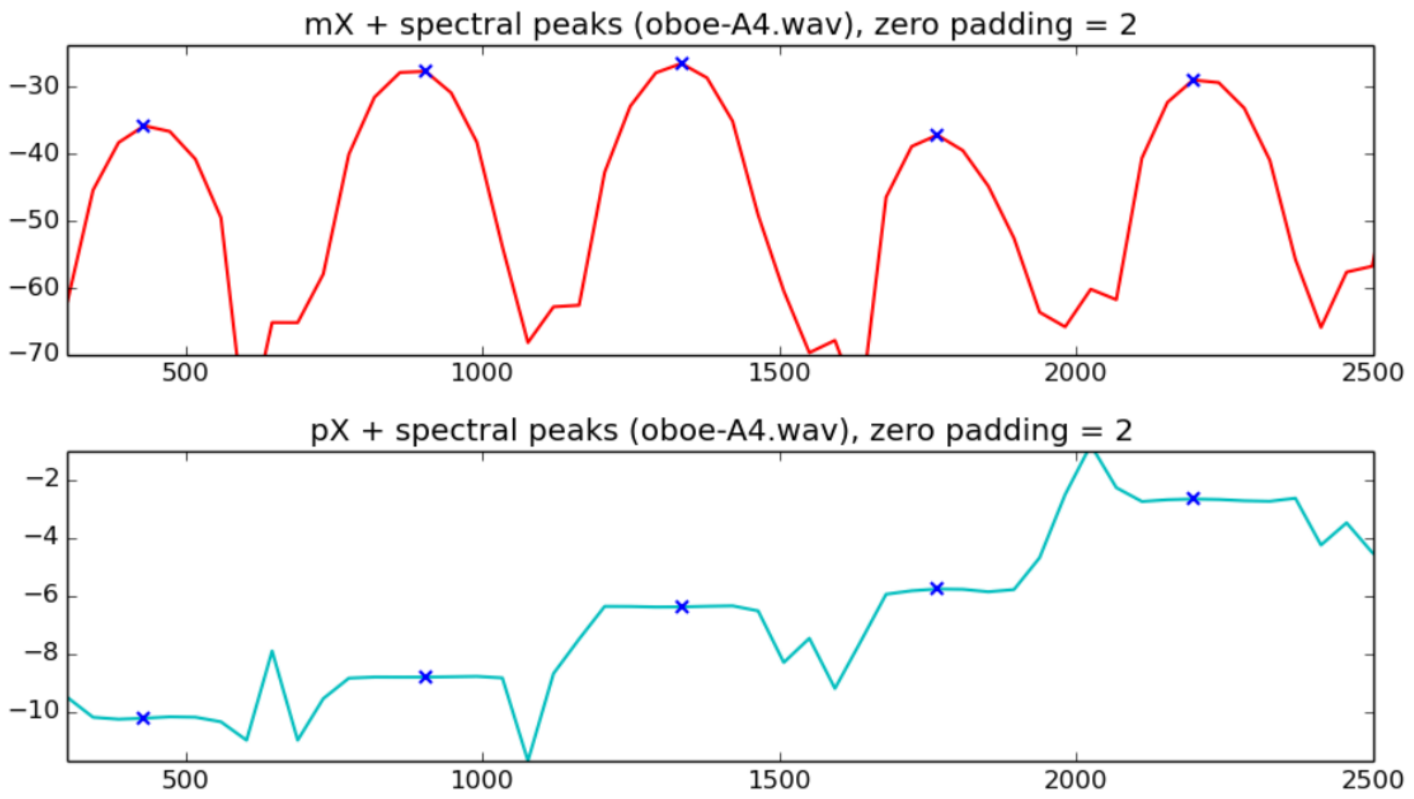
- threshold selects which maxima are kept
- the output is only a first estimate
- close components may still be merged
- phase must be read at the selected locations

Detection gives candidate bins; interpolation refines them.

Resolution problem: zero-padding helps but is not enough

Zero-padding samples the DFT more densely, but it can become computationally expensive

Peak detection with zero-padding



same peak with a denser FFT grid

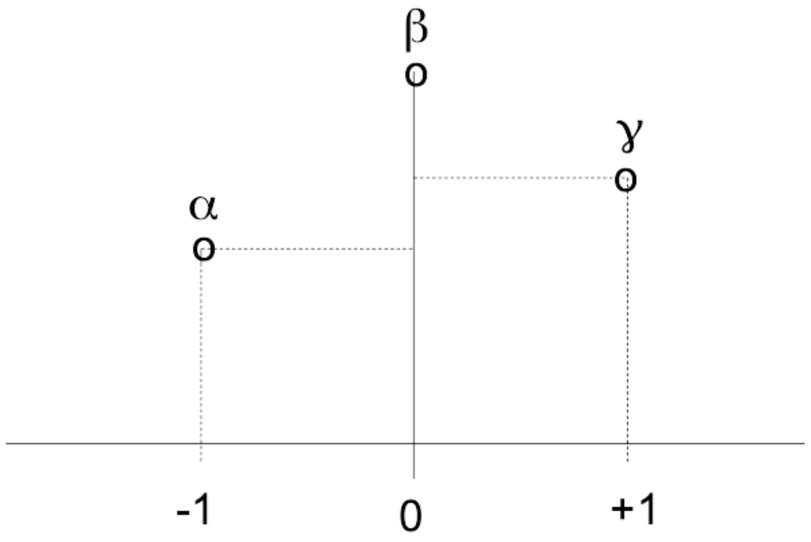
Correct interpretation

- N controls the FFT sampling grid
- M still controls the windowed frame
- more bins do not add new information
- interpolation refines the peak estimate cheaply

Parabolic interpolation refines the peak tip

Use the highest bin and its two neighbors to estimate fractional bin location and amplitude

Three-point interpolation



$\alpha = mX[k_p-1] \quad \beta = mX[k_p] \quad \gamma = mX[k_p+1]$

These three values define the local parabola around the detected maximum.

Parabolic model and outputs

$$p(n) = a(n - \hat{p})^2 + \hat{b}$$

- $\hat{p}$ is the fractional offset from the central bin
- $\hat{b}$ is the refined peak magnitude
- the refined frequency uses $\hat{k} = k_p + \hat{p}$
- phase is then evaluated at the refined position

$$\hat{p} = 1/2 \cdot (\alpha - \gamma)/(\alpha - 2\beta + \gamma)$$
$$\hat{b} = \beta - 1/4 \cdot (\alpha - \gamma)\hat{p}$$

Interpolation improves the estimate without increasing M.

From one frame to sinusoidal parameters

Each accepted peak becomes a local estimate of frequency, amplitude, and phase

Equations used in the model

$$\hat{k}_p = k_p + \hat{p}$$
$$f_p = f_s \cdot \hat{k}_p \,/\, N$$
$$A_p = \hat{m}_p$$
$$\varphi_p = \angle X[\hat{k}_p]$$

f_p: estimated frequency
A_p: estimated amplitude
ϕ_p: phase read at the refined peak

frequency, amplitude, and phase estimates from one accepted peak

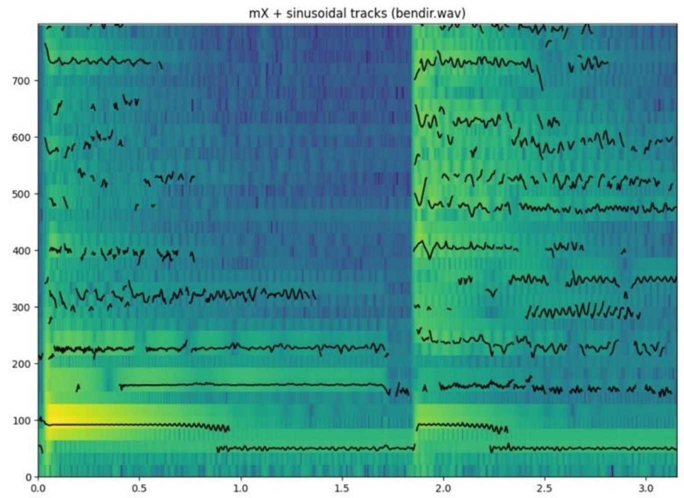
mag. peak — interp. loc. — phase at peak — store A, f, ϕ

- A single frame gives isolated estimates.
- A model needs these estimates connected over time.

From peaks to sinusoidal tracks

The model becomes time-varying only after linking peaks between frames

frame ℓ peaks → frequency continuity → amplitude continuity → phase consistency → track r through time

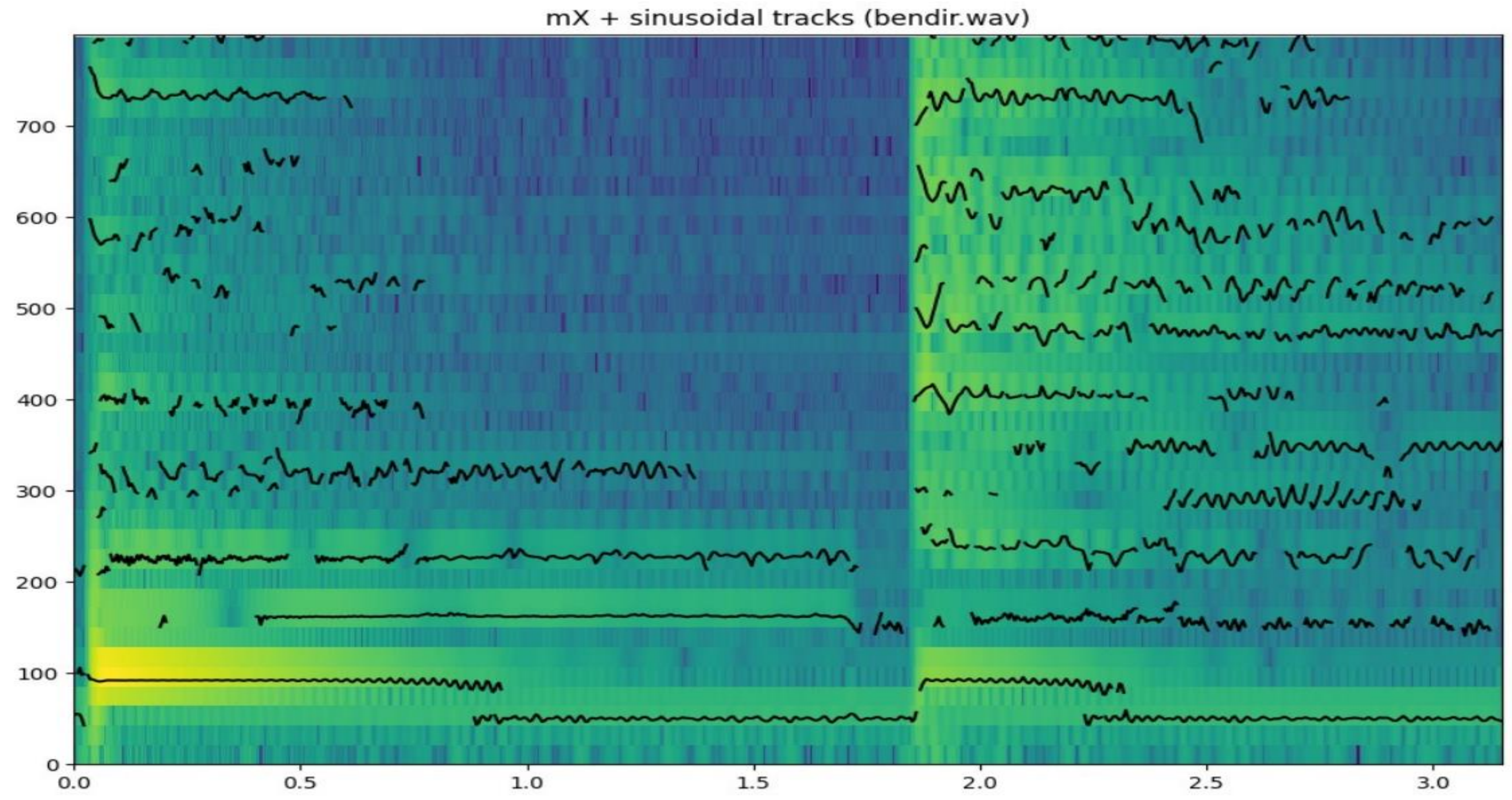


stable spectral peaks form frequency tracks over the magnitude spectrogram

EXAMPLE

Example: magnitude tracks on a spectrogram

Use the magnitude spectrogram to see where the tracked components follow strong energy ridges



bendir example: magnitude spectrogram + same track positions

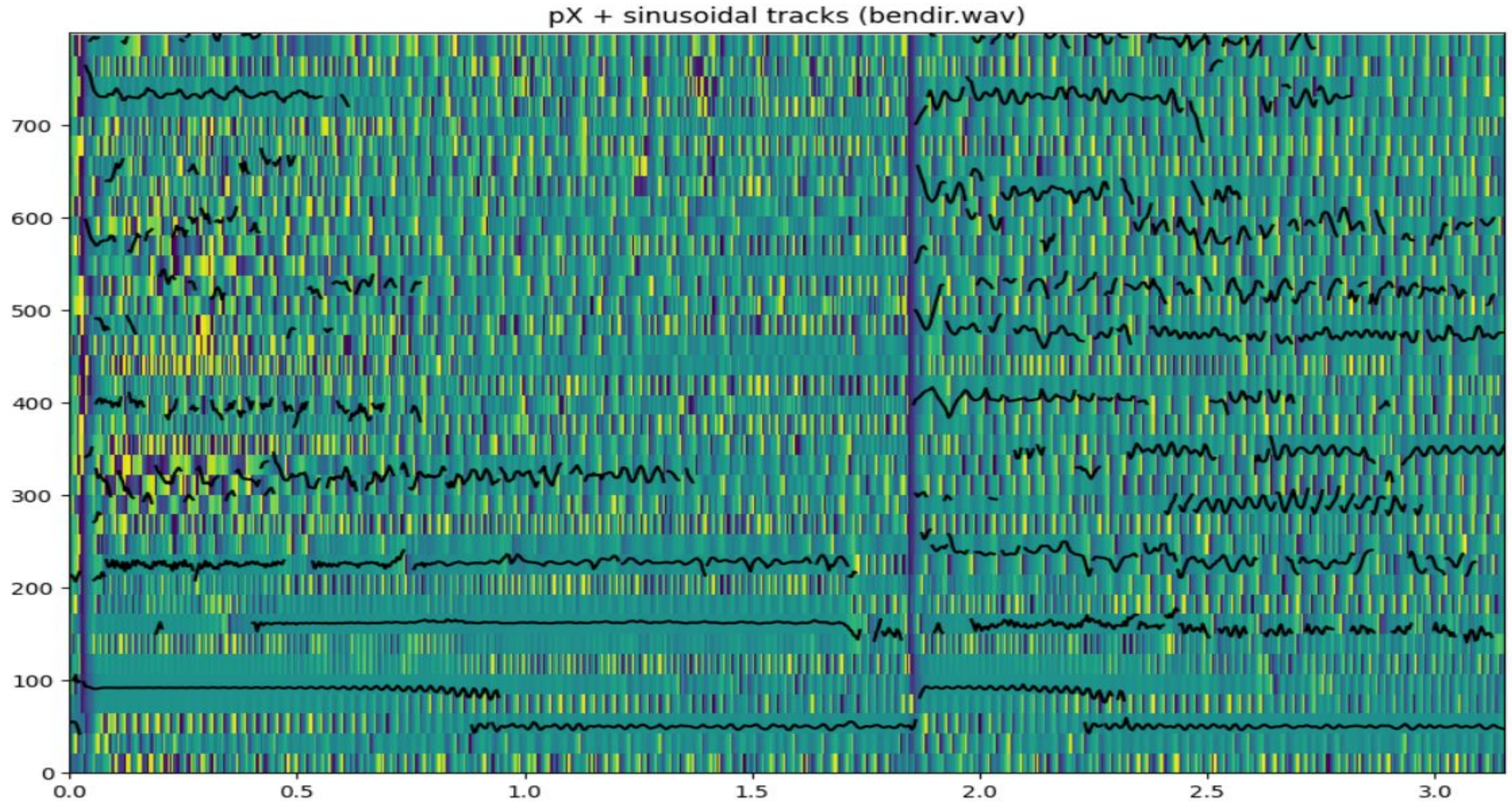
How to read it

- background color shows magnitude
- black curves are sinusoidal tracks
- strong ridges should support stable tracks
- gaps often signal weak or noisy regions

Magnitude tells us where energy is concentrated.

Phase is useful for checking sinusoidal consistency

A track is more reliable when magnitude and phase agree over time



bendir example: phase spectrogram + same track positions

Why not magnitude only?

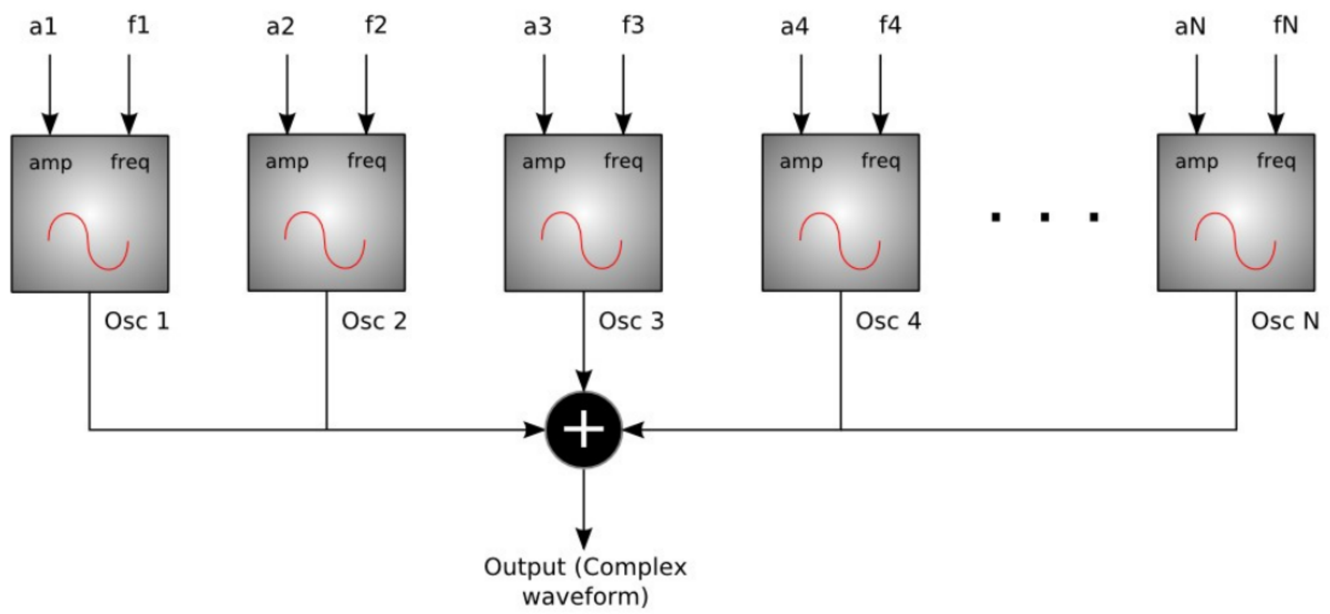
- phase carries temporal alignment
- abrupt phase behavior may reveal wrong tracks
- phase is meaningful mainly near strong peaks
- use it as a consistency check, not alone

Magnitude locates peaks; phase helps validate them.

Synthesis: parameters become sound again

After analysis, the model must generate a waveform from tracks

Sinusoidal (additive) synthesis



http://en.wikibooks.org/wiki/File:SSadditiveblock.png

Synthesis question

- for each time frame, generate the sinusoidal components
- combine them into a time-domain frame
- overlap-add frames into a continuous signal

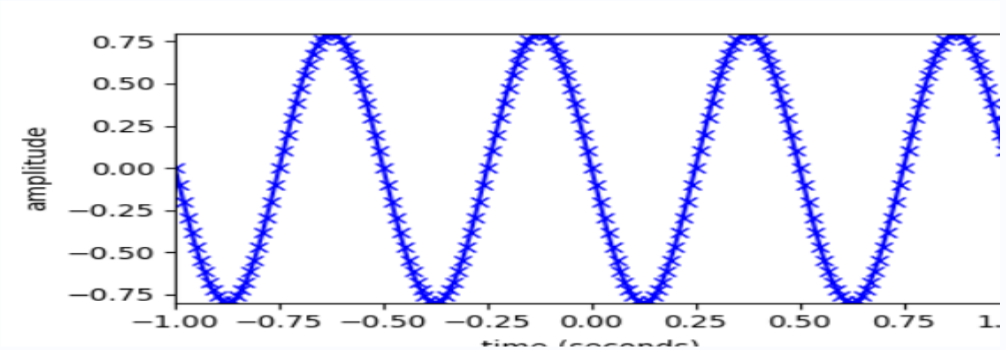
Synthesizing one sinusoid

Direct time-domain generation is simple, but spectral synthesis connects to the model

Time-domain generation

$$y[n] = A_r[n] \cos(2\pi\, f_r[n]\, n / f_s + \varphi_r)$$

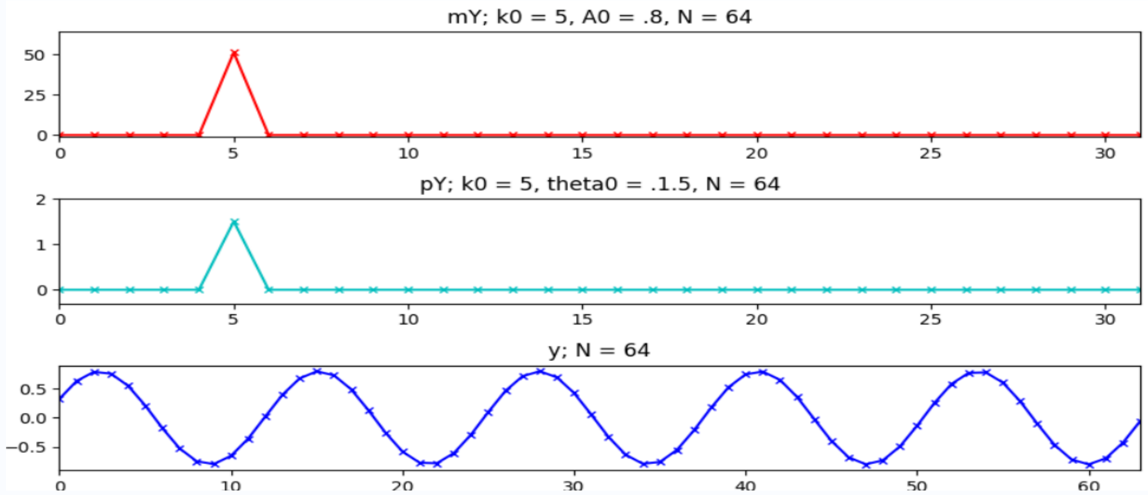
$A_r[n]$: instantaneous amplitude
$f_r[n]$: instantaneous frequency
ϕ_r: initial phase



A sinusoid can be generated directly from amplitude, frequency, and phase.

On-bin spectral synthesis

$$y[n] = \mathrm{IDFT}\{\, mY[k] \cdot e^{j pY}[k] \,\}$$
$$mY[k_0] = A_0, \quad pY[k_0] = \varphi_0$$
$$mY[k] = 0 \text{ for } k \neq k_0$$



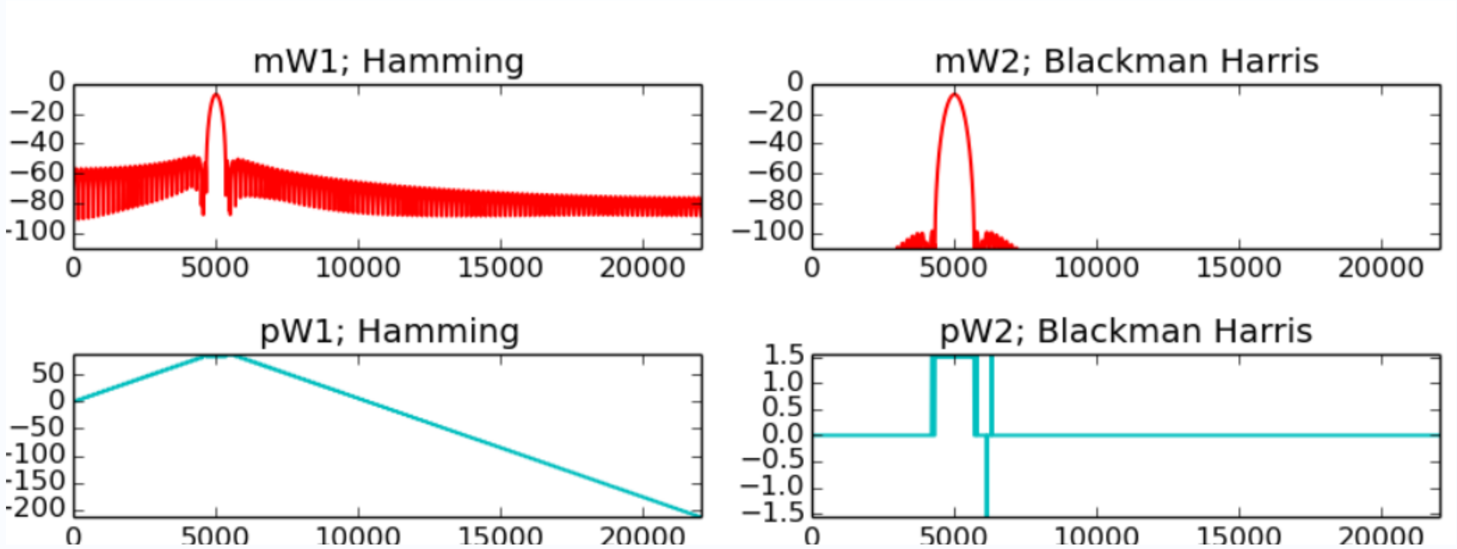
One spectral bin gives one bin-centered sinusoid after the inverse DFT.

Any frequency requires a shaped spectral lobe

Off-bin sinusoids are not represented by one isolated DFT sample

Spectral synthesis at an arbitrary frequency

$Y[k] = A_0 \cdot |W[k - k_0]| \cdot e^{j(\angle W[k-k_0] + \varphi_0)}$
$y[n] = \mathrm{IDFT}\{ Y[k] \}$
$mW[k]$, $pW[k]$: magnitude and phase of the window spectrum



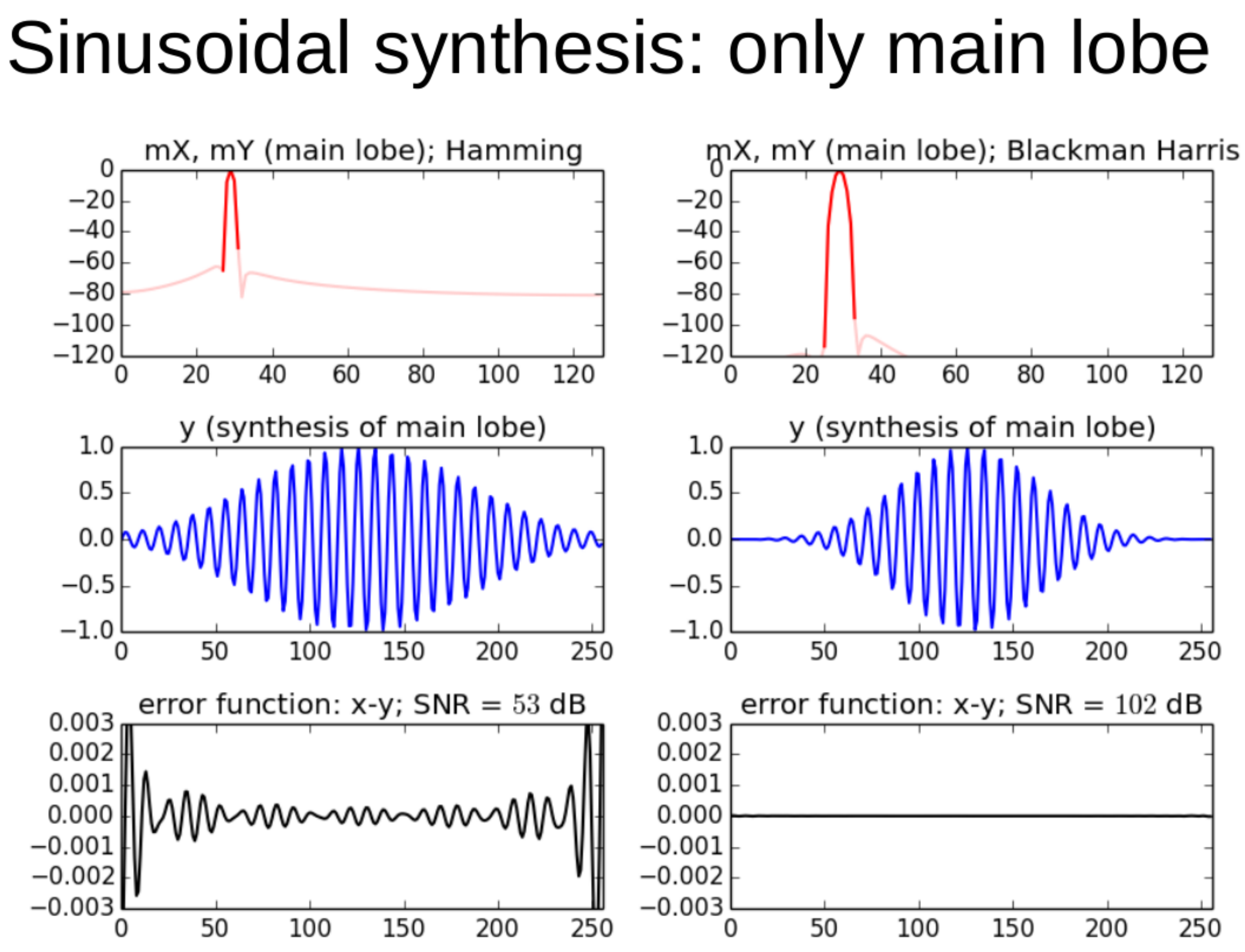
The shifted window spectrum defines magnitude and phase around the target frequency.

Synthesis rule

1. compute the window transform
2. shift it to the target frequency
3. set amplitude and phase
4. apply the inverse DFT

Main-lobe synthesis

In practice most energy is concentrated in a few bins around the peak



compare full window spectrum and main-lobe-only synthesis

Practical choice

- Hamming: first side-lobe around −42 dB
- Blackman-Harris: much lower side-lobes
- wider main lobe costs more bins
- good synthesis often uses only the main lobe

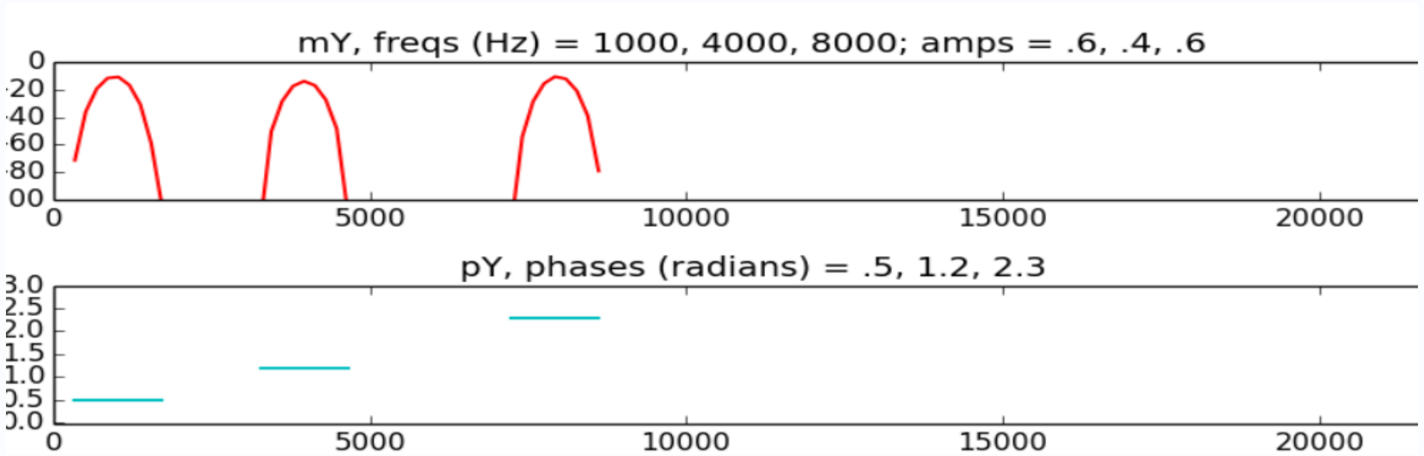
Trade-off: fewer bins versus lower leakage.

Additive synthesis in the spectrum

Several sinusoids are generated by adding several shifted main lobes

Spectrum of a sinusoidal frame

$$Y_s[k] = \Sigma_r A_r \cdot |W[k - k_r]| \cdot e^{j(\angle W[k-k_r] + \varphi_r)}$$
$$y[n] = \mathrm{IDFT}\{ Y_s[k] \}$$



Each sinusoid contributes one shifted lobe with its own amplitude and phase.

tracks → main lobes → sum $Y_s[k]$ → IFFT

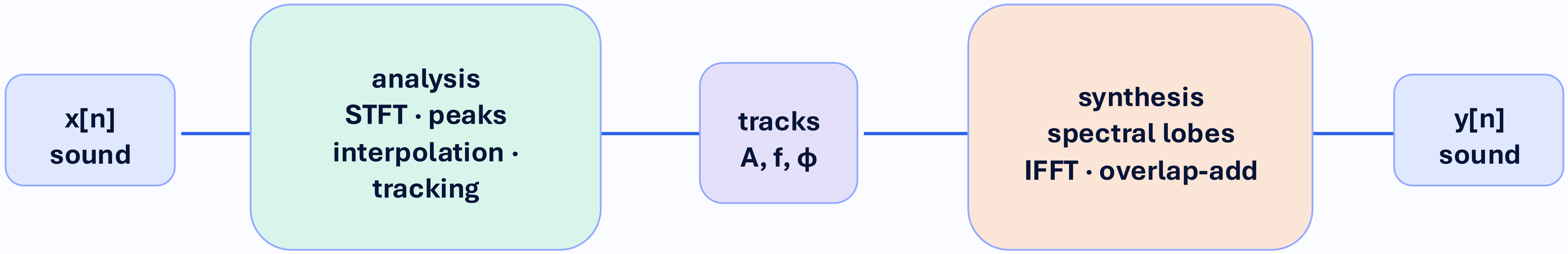
The IFFT returns a windowed frame containing all specified sinusoids.

Sinusoidal model system

Analysis detects and tracks peaks; synthesis turns tracks back into sound

Analysis side: find meaningful sinusoidal components in each frame and connect them over time.

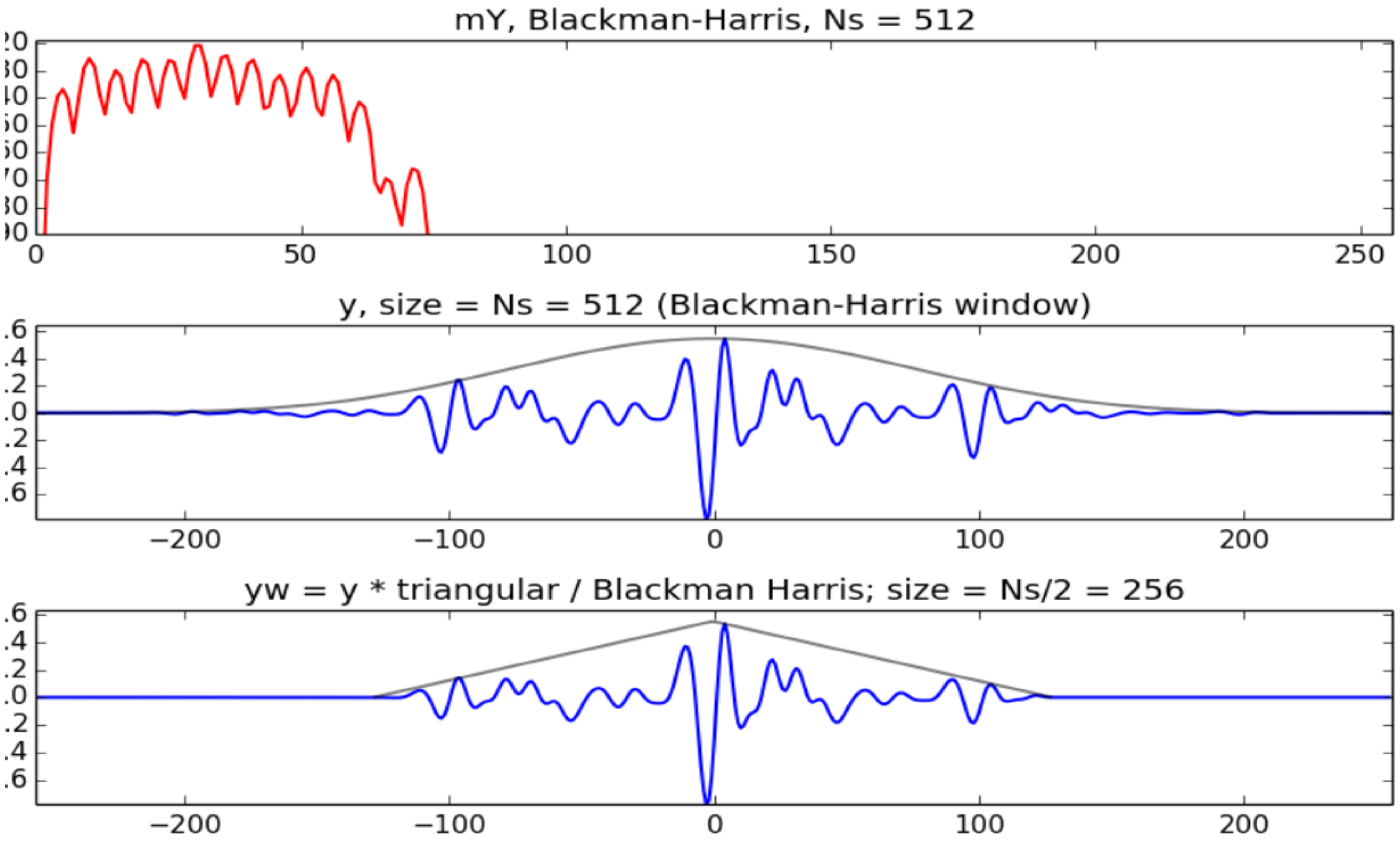
Synthesis side: turn track parameters into spectral components and reconstruct audio frames.



The tracks are the compact model: frequency, magnitude, phase, and identity through time.

Synthesis window and overlap-add

A synthesis window is needed for smooth reconstruction from overlapping frames



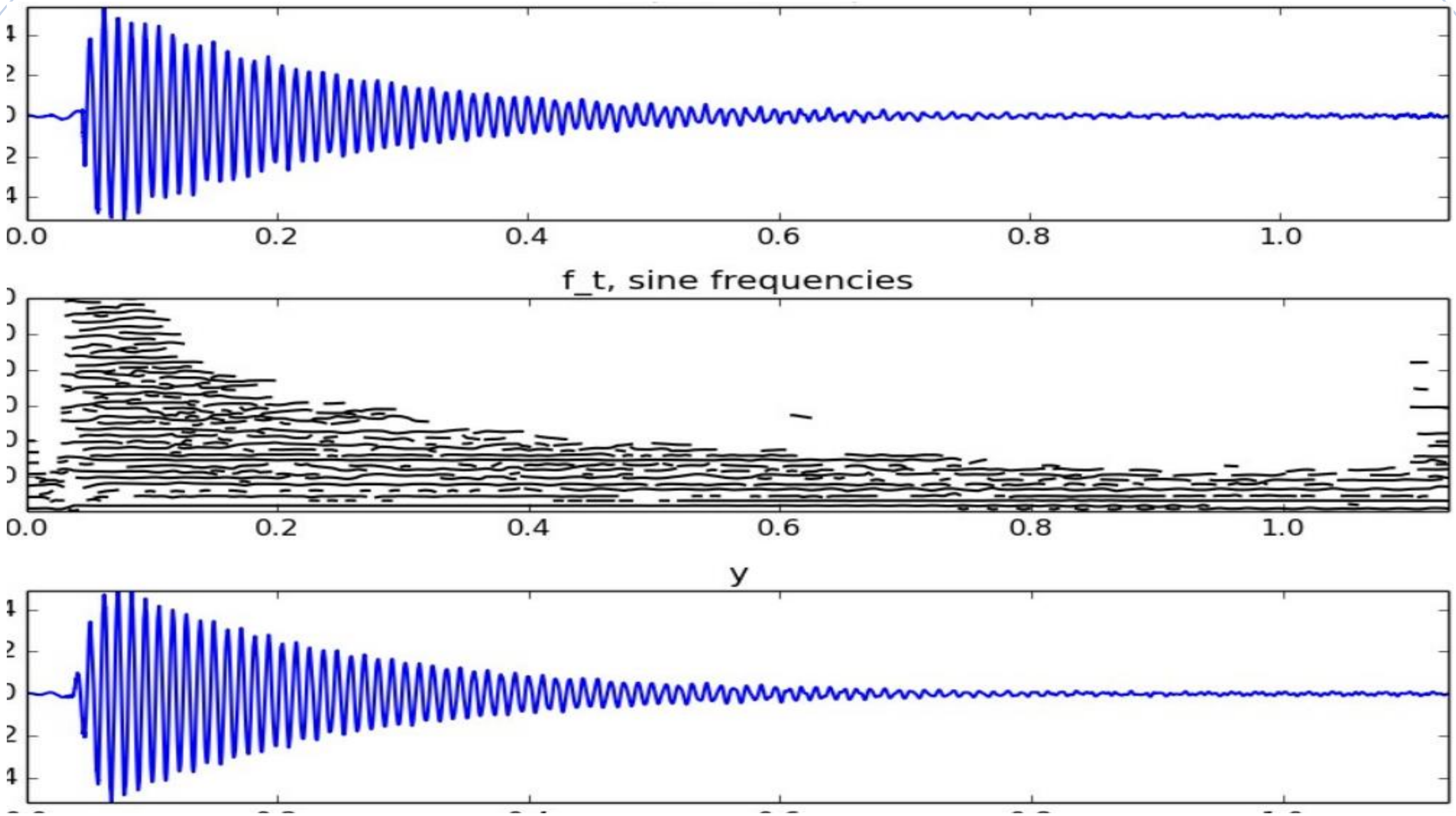
undo analysis window and apply synthesis window

25% overlap example

- generate one frame in the spectrum
- IFFT gives a time-domain segment
- apply the synthesis window
- overlap and add consecutive frames
- the hop H determines the overlap

Example: analysis/synthesis result

The final test is visual and audible: compare tracks, waveform behavior, and listening quality



input sound, sinusoidal tracks, resynthesized output

What to check

- are important tracks continuous?
- are attacks and decays preserved?
- does the output sound close enough?
- which components are missing or unstable?

Example notebooks: where each plot comes from

Use the notebooks to connect equations, arrays, parameters, tracks, and listening

examples/
peak-detection-process.ipynb
Peak detection and interpolation
threshold → local maxima → refined frequency / magnitude / phase

examples/
sine-model-examples.ipynb
Full sinusoidal-model workflow
load sound → analyze → track → synthesize → listen

examples/
sinusoidal-synthesis-examples.ipynb
Synthesis from sinusoidal parameters
A, f, ϕ tracks → spectral lobes → IFFT frames

EXAMPLES

How to read the example outputs

Do not treat plots or tracks as decorations: connect each output to a notebook and a parameter choice

1 · Peaks

From examples/peak-detection-process.ipynb: are the right components detected?

2 · Tracks

From examples/sine-model-examples.ipynb: are peaks connected without jumps?

3 · Parameters

From examples/sine-model-examples.ipynb: did M, N, H, t and window explain the result?

4 · Resynthesis

From examples/sinusoidal-synthesis-examples.ipynb: does the output match visually and by listening?

Exercise 5: Sinusoidal model

Fifth lab: estimate, track, synthesize, and diagnose sinusoids in controlled and musical sounds

What you will practice

- detect and interpolate spectral peaks
- track sinusoids across frames
- tune window, threshold, and tracking parameters
- compare plots with what you hear

How to start

Open exercises/E5-Sinusoidal-model.ipynb

Use: NumPy · Matplotlib · SciPy windows · sineModelAnal() · sineModelSynth()

1 · Estimate

single sinusoid
minimize frequency error

2 · Track

two-component chirp
choose M carefully

3 · Separate

large amplitude range
window + threshold

4 · Use phase

check track stability
avoid magnitude-only decisions

5 · Resynthesize

multiSines analysis
inspect + listen

Lab 5: change parameters deliberately

Each parameter has a visible and audible consequence

parameter	meaning	what to observe
M	window length	frequency separation vs time precision
N	FFT size	spectral sampling and peak interpolation support
t	threshold	reject weak/noisy peaks or keep faint sinusoids
maxnSines	track limit	maximum number of sinusoids per frame
minSineDur	cleaning	remove short spurious tracks
freqDevOffset / Slope	tracking tolerance	allow frequency motion between frames

What to remember from Topic 5

The sinusoidal model is a compact representation built on top of STFT analysis

1 Model
sound as time-varying sinusoids

2 Peaks
frequency, magnitude, and phase from spectra

3 Windows
M and window shape control leakage and resolution

4 Tracks
link frame-level peaks into evolving components

5 Synthesis
main lobes, IFFT, synthesis window, overlap-add

6 Validation
inspect magnitude/phase, tracks, waveform, and listening

References and credits

Use the references only when they help you implement, listen, and interpret this week's material

- Original MTG sms-tools lecture slides and examples: https://github.com/MTG/sms-tools
- MTG sms-tools-materials repository: notebooks, lecture plots, and lab materials
- examples/sine-model-examples.ipynb for analysis, tracking, and inspection
- examples/peak-detection-process.ipynb for peak detection and interpolation
- examples/sinusoidal-synthesis-examples.ipynb for spectral sinusoidal synthesis
- Serra, X. and Smith, J.O. (1990). Spectral modeling synthesis: a sound analysis/synthesis system based on a deterministic plus stochastic decomposition.